VISHWAKARMA INSTITUTES

VI

ARTIFICIAL INTELLIGENCE STUDENT FORUM

VIT CODE APEX 2.0

CODE APEX

HACK THE UPSIDE DOWN

PROBLEM STATEMENTS

DECRYPT. DESIGN. DEPLOY

VIT CODE APEX
TRACK 1 AND TRACK 2
PROBLEM STATEMENTS

TRACK 1: WEB DEVELOPMENT (COMMUDLE)

This track focuses on solving real product engineering challenges at Commudle. The problems range from infrastructure improvements to automation-driven growth systems.

1.1 PROBLEM STATEMENT 1: DOCUMENTATION MIGRATION AND SEARCH SYSTEM

Objective
Commudle currently hosts documentation in Markdown format at https://documentation. commudle.com. Build a system that:

- Migrates documentation to a sub-URL such as commudle.com/documentation.
- Preserves Markdown structure and formatting.
- Enables full-text search across all Markdown files.
- Supports scalability for future documentation expansion.
- Maintains SEO compatibility and performance standards.

The solution must integrate seamlessly seamlesslywith the existing platform architecture.

1.2 STATEMENT 2: REUSABLE REAL-TIME QUIZ MODULE OBJECTIVE

Objective

Develop a modular, plug-and-play quiz engine that can be embedded anywhere within Commudle. The system should support:

- Quiz creation interface.
- Single-question and multi-question formats.
- Multiple question types (MCQ, subjective, etc.).
- Real-time participant joining.
- Live leaderboard functionality.
- Low-latency real-time updates.
- The UI is aligned with Commudle's design system.

The architecture must be reusable, scalable, and production-ready.

1.3 STATEMENT 3: AUTOMATED SOCIAL MEDIA CONTENT GENERATION FOR FEATURE LAUNCHES

Objective

Commudle frequently launches new features but lacks an automated system for generating promotional content. Build a system that automatically generates:

- Post captions and launch copy.
- Platform-specific variations (LinkedIn, X, Instagram, etc.).
- Visual creatives (image, carousel, GIF, and short video formats).
- A ready-to-publish content package for each feature release.

The goal is to reduce turnaround time between feature launch and social media deployment.

1.4 PROBLEM STATEMENT 4: AUTOMATED FAQ GENERATION FOR FEATURE TURE PAGES

Objective

Develop a system that automatically generates structured FAQs for newly launched feature pages. The system must:

- Create context-aware FAQs based on feature descriptions.
- Optimize FAQs for SEO.
- Improve discoverability by search engines.
- Enhance AI platform comprehension of the product.

The output should be accurate, structured, and aligned with product intent.

1.5 FAKE NEWS & MISINFORMATION DETECTION

Social media platforms are flooded with fake news and misleading information.

Challenge:

Develop an AI system that detects fake news or misinformation from articles, tweets, or posts.

Expected Features:

- NLP-based content analysis
- Credibility scoring
- Source verification
- Browser extension or web app

1.6 SUSTAINABLE LIVING ASSISTANT

People want to reduce their carbon footprint but lack insights into their environmental impact.

Challenge:

Build an AI assistant that tracks and suggests sustainable lifestyle changes.

Possible Features:

- Carbon footprint tracker
- Eco-friendly product suggestions
- Energy usage prediction
- Gamified sustainability goals

TRACK 1 SUMMARY

Track 1 focuses on product-focused engineering challenges:

- Infrastructure and architecture design.
- Real-time systems.
- Automation pipelines.
- SEO and AI discoverability.

Solutions should be modular, maintainable, and deployment-ready.

TRACK 2: AI AGENTS

This track challenges participants to design and build intelligent AI agents capable of reasoning over structured and unstructured data, extracting insights, enforcing policies, and generating business artifacts.The emphasis is on explainability, modularity, and enterprise-grade scalability.

2.1 PROBLEMSTATEMENT 1: DATA DICTIONARY AGENT

Build an AI agent that can analyze relational databases and automatically generate the following:

- Schema understanding (tables, columns, keys, constraints).
- Relationship mapping and ER structure.
- Data quality metrics (nulls, completeness, freshness, consistency).
- AI-generated business context summaries.
- Human-readable data dictionaries.

Participants must handle multi-table relational databases with foreign keys and diverse data types.

Recommended Dataset (Primary)
Olist Brazilian E-Commerce Dataset
https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce License: CC BY-NC-SA 4.0

Why it fits:

- 9 interlinked relational tables.
- 100K+ orders.
- Mixed data types (timestamps, categorical, floats, text reviews, geospatial coordinates).
- Real-world schema complexity.
- Data quality variations suitable for profiling.

Alternative Datasets

Bike Store SQL Database https://www.kaggle.com/datasets/dillonmyrick/bike-store-sample-database License: CC0 (Public Domain)

Chinook Database https://github.com/lerocha/chinook-database License: MIT

Both are suitable for schema extraction and multi-database connector testing.

2.2 PROBLEM STATEMENT 2: BRD (BUSINESS REQUIREMENTS DOCUMENT) AGENT

Objective

Design an AI agent capable of extracting structured business requirements. Documents from noisy communication sources such as

- Emails.
- Meeting transcripts.
- ▪ Chat conversations.

The agent must identify anstructure the following:e:

- Functional requirements.
- Non-functional requirements.
- Stakeholders.
- Decisions and approvals.
- Timelines and milestones.
- Feature prioritization.

Recommended Dataset (Primary)

Enron Email Dataset https://www.kaggle.com/datasets/wcukierski/enron-email-dataset License: Public Domain

Why it fits:

- Approximately 500,000 real corporate emails.
- Authentic communication noise.
- Embedded project discussions and decisions.
- Stakeholder hierarchy signals.

Complementary Dataset

AMI Meeting Corpus https://huggingface.co/datasets/knkarthick/ AMI License: CC BY 4.0

- 279 meeting transcripts.
- Scenario-based design discussions.
- Includes benchmark summaries.

2.3 PROBLEM STATEMENT 3: DATA POLICY COMPLIANCE AGENT

Objective

Build an AI agent that:

- Ingests free-text policy documents (PDF format).
- Extracts enforceable rules.
- Validates those rules against structured datasets.
- Flags violations with explainable justifications.

The system must provide traceable reasoning for every compliance decision.

Recommended Dataset (Primary):
IBM Transactions for Anti-Money Laundering (AML)
https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-launderin
License: CDLA-Sharing-1.0
Why it fits:

- Synthetic but realistic financial transaction data.
- Pre-labeled compliant and violating records.
- Multiple transaction types.
- Suitable for rule-based validation.

Alternative Dataset:
PaySim Financial Dataset https://www.kaggle.com/datasets/ealaxi/paysim1 License: CC BY-SA 4.0

- 6.3 million synthetic transactions.
- Fraud-labeled records.
- Suitable for compliance simulation.

2.4 SMART AGRICULTURE USING AI

Farmers face unpredictable weather, crop diseases, and inefficient irrigation.
Challenge:
Build an AI-based farming assistant that helps farmers predict crop diseases and improve yield.
Possible Features:

- Crop disease detection via images
- Smart irrigation prediction
- Weather-based crop advisory
- Mobile-friendly interface for farmers

2.5 AI SECURITY FOR SMART SOCIETIES

Residential societies face theft, unauthorized access, and security challenges.
Challenge:
Develop an AI-powered security monitoring system that analyzes CCTV feeds and detects suspicious activities.
Expected Features:

- Intrusion detection
- Suspicious behavior detection
- Face recognition for residents
- Real-time alerts

2.6 AI FOR DISASTER MANAGEMENT

Quick response during disasters is critical but often delayed due to lack of predictive insights.
Challenge:
Build an AI platform that predicts and manages disaster response.
Possible Features:

- Flood or earthquake risk prediction
- Emergency resource allocation
- Real-time alerts
- Rescue route optimization

TRACK 3: OPEN INNOVATION

Where you can utilize and work on your own ideas.

VISHWAKARMA INSTITUTES

ARTIFICIAL INTELLIGENCE STUDENT FORUM

THANK YOU